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Research Article

Elucidating the Role of Hydrogen Bond Donor and Acceptor on Solvation in Deep Eutectic Solvents Formed by Ammonium/Phosphonium Salts and Carboxylic Acids

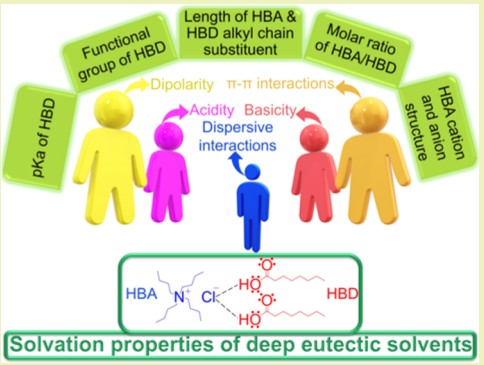
Muhammad Qamar Farooq, Gabriel A. Odugbesi, Nabeel Mujtaba Abbasi, and Jared L. Anderson*

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ABSTRACT: Deep eutectic solvents (DESs) constitute a rapidly emerging class of sustainable liquids that have been widely studied and employed in chemical separations, catalysis, and electrochemistry. The unique physicochemical and solvation properties of DESs can be highly tailored by choosing the appropriate hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD). Understanding the role of the HBA and HBD on the multiple solvation interactions in DESs is important to enable their judicious selection for particular applications. This work constitutes the first study to exploit chromatography to measure solute−solvent interactions of DESs using a wide array of known probe molecules. The constituent components of 20 DESs, formed by ammonium and phosphonium-based salts and carboxylic acids, are systematically modulated to delineate the contribution of the HBA and HBD toward individual solvation properties. Solute−solvent interactions measured in this study are used to interpret and explain the performance of DESs in desulfurization of fuels and extraction of natural products. The results from this study can be used to predict and understand the performance of DESs in various chemical processes where solvation interactions heavily influence outcomes.



KEYWORDS: *solvation interactions, gas chromatography, separations, extractions, green solvents, desulfurization, natural products*

INTRODUCTION

Deep eutectic solvents (DESs) are homogeneous mixtures formed through the combination of a hydrogen bond donor (HBD) and hydrogen bond acceptor (HBA).[1−3] The resulting compound has a lower melting point than both individual components.[4] The majority of HBAs are quaternary ammonium or phosphonium salts,[5,6] while HBDs are typically comprised of metal halides, carbohydrates, amides, alcohols, or carboxylic acids.[3,7] DESs have been applied in a number of applications such as solvents in the extraction of metals,[8,9] biocatalysis,[10−13] carbon dioxide capture,[14,15] biodiesel production,[16−18] extraction of natural compounds,[19−21] and desulfurization of fuels.[22−24] They have garnered considerable attention over the last decade and a half as potential alternatives to ionic liquids (ILs). ILs are molten organic salts comprised of a cation and anion with melting points below 100 °C. Like ILs, DESs also possess negligible vapor pressure and a broad liquid range. The physicochemical properties of DESs can generally be modulated by tailoring the chemical structure and/or the relative molar ratio of HBA/HBD.[25−27] Their hydrophobicity and hydrophilicity can also be varied by altering functional group substituents of the HBA and HBD.[5,14,28] Moreover, low-cost starting materials and solventless/purification-free synthesis make DESs more attractive than ILs in certain applications.[6,29]

To better predict the performance of DESs when they are used in chemical separations and catalysis, an understanding of their solvation interactions (i.e., hydrogen bonding, dispersion, dipolarity/polarizability, n−π, and π−π interactions) with dissolved molecules is essential. DESs have been employed as cosolvents for biocatalysis to enhance the activity and stability of lipases in aqueous reactions. Those with high hydrogen bond acidity have been observed to impart a stabilizing effect on lipases through hydrogen bond formation between the lipase and DES.[11,12] Until now, various empirical polarity scales based on solvatochromic probes have been used to characterize DESs.[17,26,30−32] Reichardt's polarity index, based on the negative solvatochromism of betaine dye 30, has been used to measure the polarity of DESs.[17,30,32−36] The normalized solvent polarity parameter (E_TN) is obtained from the dye's absorption maximum in the solvent and provides a weighted average of all solvation interactions between the probe and DES.[35] Similarly, Kamlet−Taft solvent

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parameters have been used to describe the solvation properties of DESs.[26,31,32,34−38] The hydrogen bond-donating ability (α), hydrogen bond accepting ability (β), and dipolarity/polarizability (π^*) of the solvent are determined based on the shift in absorption bands of different probes, such as betaine dye 30, 4-nitroaniline, and *N*,*N*-diethyl-4-nitroaniline.[32] Kamlet−Taft parameters describe these three interactions separately and generally provide more detail compared to the single normalized polarity parameter offered by Reichardt's dye alone. However, no single probe molecule provides a suitable measure of all solvation interactions that occur within complex solvents such as DESs.[39] Moreover, DESs comprised of acidic HBDs are known to interfere with the solvatochromic behavior of betaine dye 30 due to its zwitterionic nature.[30,32] The polarity values measured by solvatochromic probes usually fall within a narrow range and do not adequately explain experimental observations when examining DESs with different chemical makeup. For example, the hydrogen bond basicity (β value) of DESs comprised of the tetrabutylammonium bromide ($[N_{4444}^+][Br^-]$) HBA and the butanoic acid HBD were found to be almost identical (i.e., 0.81, 0.84, and 0.82) for different relative molar ratios.[31] However, for this same DES, a 10-fold increase in extraction efficiency of cynaropicrin was observed when the HBA/HBD molar ratio was varied from 1:1 to 1:2.[40]

Limitations of solvatochromic methods necessitate approaches that use a broad array of probe molecules to measure solute−solvent interactions. Inverse gas chromatography (IGC) employs a solvent as the chromatographic stationary phase, and the extent to which known probe molecules interact provides an indirect measure of the strength of individual solvation interactions. This approach requires very small amounts of the DES (15−20 mg) and permits the solvation properties to be examined as a function of temperature.[41] Individual solvation interactions are determined using a linear free energy relationship, such as the Abraham solvation parameter model (eq 1), which describes the contribution of each individual solvation interaction based on the chromatographic retention of probe molecules.[42,43]

$$\text{Log } k = c + eE + sS + aA + bB + lL \quad (1)$$

As shown in eq 1, k refers to the retention factor of each probe molecule for the DES stationary phase at a specific temperature. The retention factor is calculated chromatographically by measuring the retention time of each solute as well as the column dead volume. The solute descriptors (E, S, A, B, and L) have been experimentally determined by Abraham[42] and are defined as: E, excess molar refraction calculated from the solute's refractive index; S, solute dipolarity/polarizability; A, solute hydrogen bond acidity; B, solute hydrogen bond basicity; and L, solute gas−hexadecane partition coefficient at 298 K.[42] Retention factors and solute descriptors are used to measure the system constants (c, e, s, a, b, l) that characterize the multiple solvation interactions of the solvent. The system constants include: e is the ability of the solvent to interact with π- and n-electrons of the solute; s describes the dipolarity/polarizability of the solvent; a defines the hydrogen bond basicity (i.e., interaction of the basic solvent with acidic solutes); b is the measure of the hydrogen bond acidity; and l describes dispersive interactions. This model has been utilized to characterize the solvation properties for various classes of solvents.[44−46]

In this study, an extensive series of 20 DESs formed by ammonium and phosphonium-based salts and carboxylic acids are characterized by IGC. This class of DESs has been employed in a number of interdisciplinary fields of chemistry,[8,13,47,48] but efforts toward fully characterizing their solvation properties have been met with limited success. This represents the first study to measure chromatographically solute−solvent interactions of DESs. The breadth of DES combinations was chosen to examine the effect of the following features on solvation interactions: (a) molar ratio of HBA/HBD, (b) length of HBA and HBD alkyl chain substituent, (c) structure and combination of HBA cation and anion, and (d) pK_a of HBD. An enhanced understanding of HBA and HBD interactions and their overall effect on solvation interactions aids in the development of a molecular model that can ultimately be used to interpret and possibly predict the performance of DESs in applications such as chemical separations and catalysis.

EXPERIMENTAL SECTION

Materials. Compounds used for the preparation of DESs including tetraethylammonium chloride ($[N_{2222}^+][Cl^-]$, >98%), tetrabutylammonium chloride ($[N_{4444}^+][Cl^-]$, >97%), $[N_{4444}^+][Br^-]$ (>99%), allyl(triphenyl)phosphonium bromide ($[P_{Al(Ph)3}^+][Br^-]$, 99%), tetrabutylphosphonium chloride ($[P_{4444}^+][Cl^-]$, 96%), hexanoic acid (HA, >98%), octanoic acid (OA, >99%), *para*-toluenesulfonic acid monohydrate (TSA, >98.5%), levulinic acid (LvA, 98%), and benzenesulfonic acid (BSA, 98%) were purchased from MilliporeSigma (St. Louis, MO). Tetrapropylammonium chloride ($[N_{3333}^+][Cl^-]$, >97%) was purchased from Tokyo Chemical Industry (Portland, OR), and trihexyl(tetradecyl)phosphonium chloride ($[P_{66614}^+][Cl^-]$, >93%) was purchased from Strem Chemical (Newburyport, MA). Malonic acid (MA, 99%) was purchased from Acros Organics (Morris Plains, NJ), and L-lactic acid (LcA, 98%) was purchased from Alfa Aesar (Ward Hill, MA).

The following compounds were used for IGC measurements. Butyraldehyde (99%), 1-chlorobutane (99%), ethyl acetate (99.5%), methyl caproate (99%), naphthalene (99%), cyclopentanol (99%), nitromethane (99%), and 2-nitrophenol (99%) were purchased from Acros Organics. Bromoethane (98%) was purchased from Alfa Aesar, and ethyl benzene was purchased from Eastman Kodak Company (Rochester, NJ). Acetic acid (99.9%), *N*,*N*-dimethylformamide (*N*,*N*-DMF, 99.9%), 1-hexanol (98%), cycloheptanol (98%), and toluene (99.8%) were purchased from Fisher Scientific (Pittsburgh, PA). 2-Chloroaniline (98%), *p*-cresol (99%), *o*-xylene (97%), *p*-xylene (99.5%), methyl acetate (98%), phenylethyne (98%), and 1-bromohexane (98%) were purchased from Fluka (Steinheim, Germany). Benzaldehyde (99%), 5-bromoacenapthene (90%), 2-nitronapthalene (85%), 1-chlorohexane (99%), 1-chlorooctane (99%), cyclohexanol (99%), cyclohexanone (99.8%), 1-iodobutane (99%), iodoethane (99%), 1-nitropropane (98%), octylaldehyde (99%), 1-pentanol (99%), 2-pentanone (99%), propionitrile (99%), 1-decanol (99%), acetophenone (99%), aniline (99.5%), benzonitrile (99%), benzyl alcohol (99%), 1-bromooctane (99%), 2-butanol (98%), 1-butanol (99.8%), 1,2-dichlorobenzene (99%), dichloromethane (99.8%), 1,4-dioxane (99.5%), 1-octanol (99%), phenol (99%), pyridine (99%), pyrrole (98%), *m*-xylene (99.5%), 1-propanol (99.9%) 2-propanol (99.9%), methanol (99%), ethanol (99%), benzylamine (99%), benzamide (99%), and propionic acid (99%) were purchased from MilliporeSigma (St. Louis, MO).

Deuterated dimethyl sulfoxide (DMSO-d_6) and chloroform ($CDCl_3$) were obtained from Cambridge Isotope Laboratories (Andover, MA). Deactivated capillary (5 m × 250 μm) was obtained from MEGA (Legnano, MI, Italy) and used for the preparation of chromatographic columns.

Methods. *Preparation of DESs.* DESs examined in this study (Figure 1) have been previously reported and their preparation

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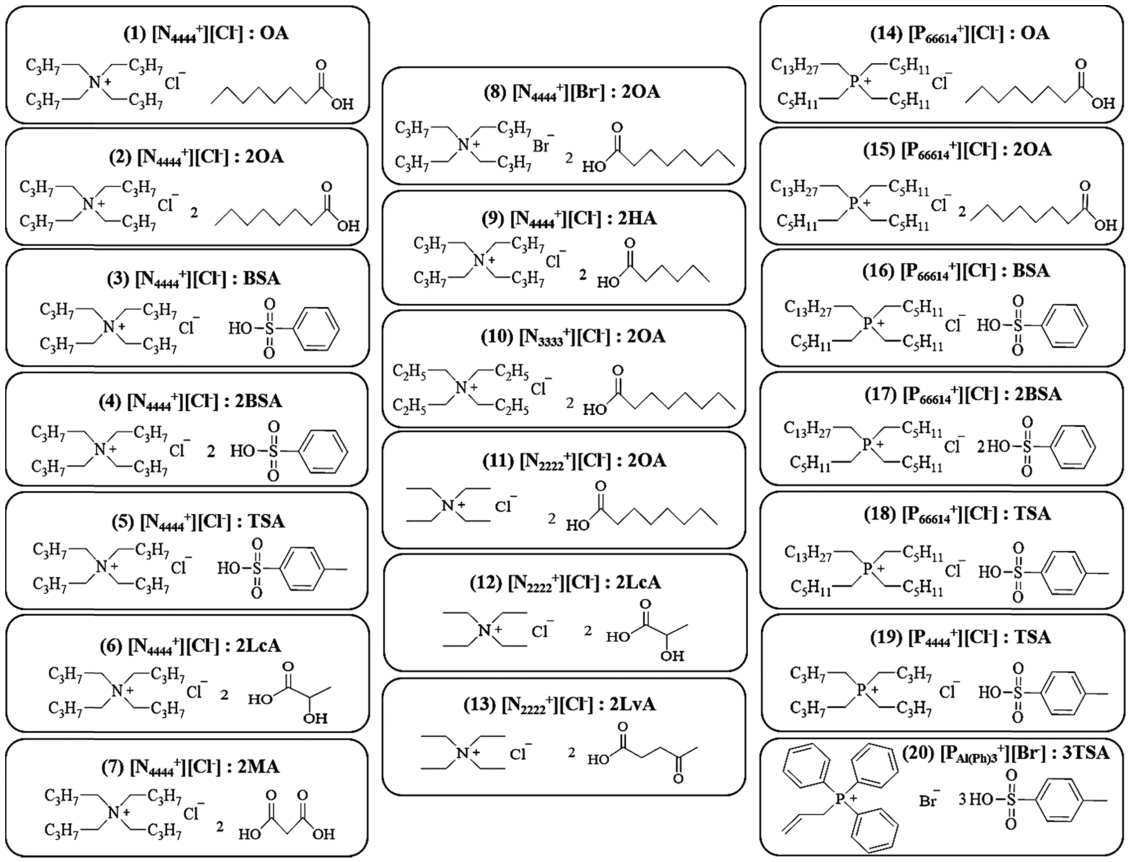


Figure 1. Chemical structures, relative ratios of HBA/HBD, abbreviations, and numbering scheme used for the 20 DESs evaluated in this study.

involved the use of similar protocols.[8,18,31,32,48] First, equimolar amounts of the HBD and HBA were weighed in a 20 mL vial containing a magnetic stirrer. The vial was then heated for 3 h at 60 °C, after which a uniform and homogeneous mixture of DES formed. The DES was placed to dry in a vacuum oven at room temperature for 2 days. Water content of all DESs, provided in Table S1, was measured by a Metrohm 831 Karl Fischer coulometric titrator. Melting point and/or glass transition temperatures for all DESs are provided in Table S2, and a representative phase diagram for the $[N_{4444}^+][Cl^-]$:OA DES is shown in Figure S1. Differential scanning calorimetry (DSC) measurements were performed using a DSC Q2000 calorimeter (TA Instruments). All samples were cooled from 40 to −120 °C at a rate of 20 K/min and then heated to 80 °C at a rate of 20 K/min. DESs examined in this study were characterized by ^{13}C and ^{1}H NMR, and all spectra are provided in the Supporting Information.

Chromatographic Column Preparation Using DES as Stationary Phase. All gas chromatography (GC) columns were prepared by coating a thin layer of DES on the inner wall of 5 m deactivated fused silica capillaries using the static coating method. All DESs were dissolved in dichloromethane to prepare a coating solution (0.45% w/v) that produced an approximate film thickness of 0.28 μm.[49] The coated columns were conditioned at 60 °C for 30 min under a constant flow of helium. Column efficiencies were determined using naphthalene at 60 °C, and all columns possessed efficiencies above 2200 plates/m.

For the IGC study, all probe molecules were dissolved in dichloromethane at a concentration of 1 mg/mL. A total of 52 probe molecules, shown in Table S3, were used in this study and were injected individually at 40, 50, and 60 °C. Analytes possessing lower boiling points exhibited little to no retention on the stationary phase, while other analytes were retained more strongly. Therefore, not all probe molecules were subjected to the solvation parameter model at some temperatures.

GC measurements were carried out on an Agilent Technologies 7890B gas chromatograph employing a flame ionization detector (GC-FID). Helium was used as a carrier gas at a flow rate of 1 mL/min. The injector and detector temperatures were held at 150 °C using a split ratio of 20:1, and an injection volume of 1 μL was used in all experiments. Hydrogen was utilized as the makeup gas at a flow rate of 30 mL/min, while the airflow was held at 400 mL/min. Propane was used to measure the dead volume of each column. Multiple linear regression analysis and statistical calculations were performed using the program Analyze-it (Microsoft). Figure S2 demonstrates a typical regression line consisting of all probe molecules with a correlation coefficient (R) value of 0.99.

RESULTS AND DISCUSSION

This study examines DESs comprised of ammonium and phosphonium-based salts as HBAs and a series of carboxylic acids as HBDs with pK_a values ranging from −6.7 to 4.86. The chemical structures of the HBAs and HBDs, as well as abbreviations and numbering of DESs, are shown in Figure 1. DESs **1**−**19** were prepared using two different molar ratios of HBA/HBD (i.e., 1:1 and 1:2), and DES **20** was prepared using a 1:3 molar ratio to study the effect of this variable on the overall solvation characteristics.

At the onset of this study, several challenges were encountered in measuring solute−solvent interactions of DESs by IGC. To obtain chromatographic columns with high separation efficiency, it is critical that a uniform thin film of solvent be maintained on the inner wall of the capillary. Furthermore, the thin film must retain its integrity and should not flow when subjected to varying temperatures. When injector/detector temperatures of 250 °C and oven temperatures exceeding 80 °C were employed in initial GC

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separations, it was observed that the retention times of probe molecules varied significantly, indicating disruption to the DES stationary phase. Retention times were found to be reproducible on newly coated capillaries when the injector/detector and oven temperatures were maintained at 150 and 60 °C, respectively, under a constant flow of dry carrier gas. These temperatures permitted the probe molecules to be volatilized for chromatographic analysis while preventing loss of the DES stationary phase. The system constants of all DESs were examined at 40, 50, and 60 °C using a diverse and broad range of probe molecules (see Table S3). The retention characteristics of each probe were measured at three different temperatures to ensure the soundness of the solvation models and to examine the variation in solvation interactions with temperature. Figure 2 shows the GC separation of alcohols and haloalkanes on three ammonium-based DES stationary phases. The structural features and composition of each DES resulted in varying solvation interactions and different separation behavior for the 10 probe molecules.

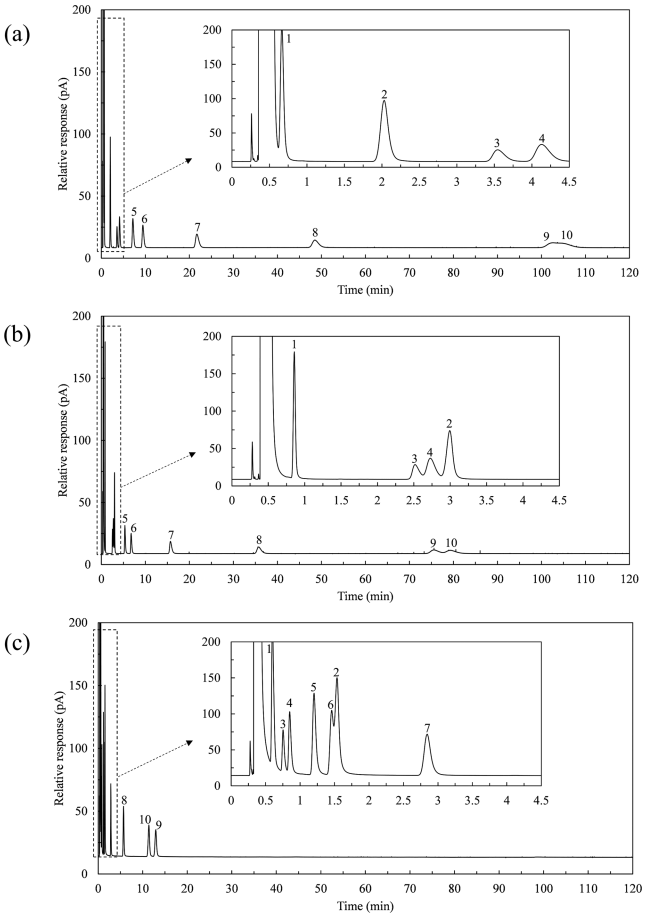


Figure 2. Chromatographic separation of alcohols and haloalkanes on (a) $[N_{4444}^+][Cl^-]$:OA, (b) $[N_{4444}^+][Cl^-]$:2OA, and (c) $[N_{4444}^+][Cl^-]$:2LcA DES stationary phases at 60 °C. The inset within each chromatogram shows the first 4.5 min of the separation so that all chromatographic peaks can be observed. Analytes: 1, chlorohexane; 2, chlorooctane; 3, methanol; 4, ethanol; 5, 2-butanol; 6, propanol; 7, butanol; 8, pentanol; 9, cyclohexanol; and 10, hexanol.

Values for the system constants at the three temperatures are shown in Tables 1 and 2 for the ammonium and phosphonium-based DESs, respectively. All of the developed models are statistically sound based on the correlation coefficients of the multiple linear regression fit and the magnitude of the Fisher *F*-statistics. As expected, the majority of system constants exhibited a smooth drop with increasing temperature. $[N_{4444}^+][Cl^-]$ and $[P_{66614}^+][Cl^-]$ ILs were used as HBAs to form many of the DESs in this study, and their system constants are also provided.

Comparison of Solvation Interactions for Ammonium and Phosphonium-Based DESs. Among the 13 ammonium-based DESs studied (DESs **1**−**13**, see Table 1), they were all found to exhibit strong hydrogen bond basicity (a), dipolarity (s), and dispersive-type interactions (l). Dipolar interactions ranged from 1.97 to 2.52 at 50 °C, and all ammonium-based DESs were less dipolar than the $[N_{4444}^+][Cl^-]$ IL. Dispersion interactions at 50 °C ranged from 0.62 to 0.73 for DESs **1**−**11** and were considerably lower for the $[N_{2222}^+][Cl^-]$:2LcA and $[N_{2222}^+][Cl^-]$:2LvA DESs. Phosphonium-based DESs (**14**−**20**) examined in this study also possessed strong hydrogen bond basicity, dipolarity, and dispersive-type interactions. Dipolar interactions for these DESs ranged from 1.86 to 2.56 at 50 °C and generally exhibited similar (DESs **14**−**17**, **20**) or higher (DESs **18**−**19**) dipolarity than the $[P_{66614}^+][Cl^-]$ IL. Overall, phosphonium-based DESs exhibited considerably higher dispersive-type interactions compared to the ammonium-based DESs.

To classify the 20 DESs examined in this study based on two dominant interactions, their hydrogen bond basicity and dispersion interaction values from the model were plotted. For comparison purposes, the $[N_{4444}^+][Cl^-]$ and $[P_{66614}^+][Cl^-]$ ILs are also included. Using k-means clustering (Figure S3), six groupings were attained. As shown in Figure 3, DESs in group A consist of the $[N_{4444}^+][Cl^-]$ and $[P_{66614}^+][Cl^-]$ HBDs and higher pK_a HBAs (OA and HA). Group B is made up exclusively of the $[P_{66614}^+][Cl^-]$ HBA with different molar ratios of the BSA HBD. DESs generally composed of HBAs with moderate to long alkyl chains ($[N_{4444}^+][Cl^-]$, $[N_{4444}^+][Br^-]$, ($[N_{3333}^+][Cl^-]$), $[P_{4444}^+][Cl^-]$, $[P_{66614}^+][Cl^-]$), and OA and TSA HBAs comprise group C. DESs in group D include the $[N_{4444}^+][Cl^-]$ and $[P_{Al(Ph)3}^+][Cl^-]$ HBAs and low pK_a HBDs (LcA, MA, TSA, and BSA). Groups E and F include the $[N_{2222}^+][Cl^-]$:2LvA and $[N_{2222}^+][Cl^-]$:2LcA DESs, respectively. Figure 3 provides a chemically useful means to differentiate the solvents from one another for use in specific applications and demonstrates the important role that the HBA and HBD have in modulating these solvation interactions.

In a previous study comparing the solute−solvent interactions of ILs measured by the solvation parameter model and Kamlet−Taft study,[50] it was proposed that the solvation parameter model tended to report lower hydrogen bond acidity when evaluating ILs with highly basic anions whereas Reichardt dye interacts exclusively with the cation. According to Tables 1 and 2, this is clearly the case for the hydrogen bond acidity parameter, where moderate to high negative values are observed. However, two previous studies have reported contradictory Kamlet−Taft parameter data for the $[N_{4444}^+][Cl^-]$:2OA and $[N_{4444}^+][Cl^-]$:2decanoic acid DESs. Florindo et al.[32] reported a hydrogen bond-donating ability (α) of 1.41 and 1.36 and hydrogen bond acceptor ability (β) of 0.99 and 0.97 for the $[N_{4444}^+][Cl^-]$:2OA and $[N_{4444}^+][Cl^-]$:2decanoic acid DESs, respectively, while Teles et al.[31] reported α of 0.84 and 0.85 and β of 1.19 and 1.28, respectively, for the same DESs. The following sections discuss the effects of hydrogen bond acceptor and hydrogen bond donor on overall DES solvation characteristics.

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Table 1. System Constants for 13 Ammonium-Based DESs with Varying HBAs and HBDs Obtained from the Solvation Parameter Model[a]

			system constants								
no.	DES	temp. (°C)	*c*	*e*	*s*	*a*	*b*	*l*	*n*[b]	R^2[c]	*F*[d]
1	$[N_{4444}^+][Cl^-]$:OA	40	−3.38 (0.10)	−0.14 (0.09)	2.67 (0.12)	8.11 (0.18)	−0.43 (0.14)	0.76 (0.02)	39	0.99	774
		50	−3.31 (0.09)	−0.10 (0.08)	2.52 (0.11)	7.60 (0.16)	−0.40 (0.14)	0.70 (0.02)	39	0.99	803
		60	−3.35 (0.09)	−0.13 (0.08)	2.50 (0.11)	7.41 (0.16)	−0.53 (0.13)	0.68 (0.02)	39	0.99	776
2	$[N_{4444}^+][Cl^-]$:2OA	40	−3.16 (0.08)	−0.18 (0.08)	2.41 (0.10)	7.13 (0.15)	−0.36 (0.13)	0.77 (0.02)	39	0.99	884
		50	−3.18 (0.07)	−0.16 (0.07)	2.34 (0.10)	6.80 (0.14)	−0.38 (0.12)	0.73 (0.02)	39	0.99	960
		60	−3.19 (0.07)	−0.14 (0.06)	2.26 (0.09)	6.48 (0.13)	−0.38 (0.11)	0.69 (0.02)	39	0.99	1037
3	$[N_{4444}^+][Cl^-]$:BSA	40	−3.10 (0.07)	−0.20 (0.07)	2.48 (0.08)	5.34 (0.12)	−0.38 (0.11)	0.72 (0.02)	41	0.99	1015
		50	−3.12 (0.06)	−0.20 (0.06)	2.46 (0.08)	5.23 (0.11)	−0.44 (0.10)	0.68 (0.01)	41	0.99	1134
		60	−3.19 (0.06)	−0.17 (0.06)	2.38 (0.08)	4.94 (0.11)	−0.38 (0.10)	0.66 (0.01)	41	0.99	1029
4	$[N_{4444}^+][Cl^-]$:2BSA	40	−3.23 (0.06)	−0.14 (0.06)	2.52 (0.08)	4.54 (0.13)	−0.12 (0.11)	0.70 (0.01)	40	0.99	1111
		50	−3.19 (0.06)	−0.11 (0.06)	2.40 (0.07)	4.29 (0.11)	−0.17 (0.10)	0.65 (0.01)	40	0.99	1225
		60	−3.22 (0.05)	−0.08 (0.05)	2.31 (0.06)	4.00 (0.10)	−0.13 (0.09)	0.62 (0.01)	40	0.99	1337
5	$[N_{4444}^+][Cl^-]$:TSA	40	−3.20 (0.07)	−0.32 (0.07)	2.58 (0.09)	5.99 (0.14)	−0.59 (0.12)	0.78 (0.02)	38	0.99	968
		50	−3.20 (0.07)	−0.28 (0.07)	2.49 (0.09)	5.66 (0.13)	−0.58 (0.11)	0.73 (0.02)	38	0.99	988
		60	−3.16 (0.07)	−0.24 (0.07)	2.42 (0.09)	5.50 (0.13)	−0.60 (0.11)	0.68 (0.02)	38	0.99	892
6	$[N_{4444}^+][Cl^-]$:2LcA	40	−2.93 (0.07)	0 (0)	2.20 (0.09)	5.03 (0.14)	0 (0)	0.68 (0.02)	38	0.99	832
		50	−3.06 (0.05)	−0.06 (0.05)	2.16 (0.06)	4.64 (0.10)	0 (0)	0.66 (0.01)	38	0.99	1351
		60	−3.02 (0.05)	0 (0)	2.08 (0.06)	4.25 (0.09)	0 (0)	0.61 (0.01)	38	0.99	1467
7	$[N_{4444}^+][Cl^-]$:2MA	40	−3.09 (0.06)	0 (0)	2.31 (0.07)	4.78 (0.11)	0 (0)	0.66 (0.01)	39	0.99	1302
		50	−3.11 (0.05)	0 (0)	2.24 (0.06)	4.41 (0.10)	0 (0)	0.62 (0.01)	38	0.99	1330
		60	−3.15 (0.05)	0 (0)	2.17 (0.06)	4.04 (0.10)	0.09 (0.08)	0.58 (0.01)	39	0.99	1422
8	$[N_{4444}^+][Br^-]$:2OA	40	−3.18 (0.07)	−0.21 (0.07)	2.24 (0.09)	5.88 (0.13)	−0.46 (0.11)	0.79 (0.02)	36	0.99	1026
		50	−3.13 (0.07)	−0.17 (0.07)	2.15 (0.09)	5.57 (0.13)	−0.42 (0.11)	0.73 (0.02)	35	0.99	914
		60	−3.13 (0.05)	−0.19 (0.05)	2.05 (0.06)	5.05 (0.09)	−0.43 (0.08)	0.71 (0.01)	36	0.99	1621
9	$[N_{4444}^+][Cl^-]$:2HA	40	−3.17 (0.09)	−0.13 (0.08)	2.36 (0.11)	7.00 (0.17)	−0.32 (0.14)	0.75 (0.02)	38	0.99	739
		50	−3.14 (0.08)	−0.14 (0.07)	2.25 (0.10)	6.44 (0.15)	−0.32 (0.12)	0.71 (0.02)	38	0.99	840
		60	−3.11 (0.07)	−0.14 (0.07)	2.17 (0.09)	5.92 (0.13)	−0.36 (0.11)	0.67 (0.02)	38	0.99	849
10	$[N_{3333}^+][Cl^-]$:2OA	40	−3.04 (0.08)	−0.16 (0.08)	2.15 (0.10)	5.96 (0.16)	−0.47 (0.13)	0.77 (0.02)	37	0.99	705
		50	−3.06 (0.07)	−0.13 (0.07)	2.05 (0.09)	5.43 (0.14)	−0.45 (0.11)	0.75 (0.01)	37	0.99	822
		60	−2.99 (0.07)	−0.22 (0.07)	1.91 (0.09)	5.09 (0.14)	−0.55 (0.12)	0.73 (0.01)	37	0.99	658
11	$[N_{2222}^+][Cl^-]$:2OA	40	−3.32 (0.06)	−0.10 (0.06)	2.04 (0.08)	5.30 (0.12)	−0.29 (0.10)	0.78 (0.02)	38	0.99	1155
		50	−3.22 (0.06)	−0.08 (0.06)	1.97 (0.07)	5.02 (0.11)	−0.23 (0.09)	0.72 (0.01)	37	0.99	1204
		60	−3.07 (0.05)	0 (0)	1.93 (0.07)	4.86 (0.10)	−0.26 (0.09)	0.66 (0.01)	38	0.99	1306
12	$[N_{2222}^+][Cl^-]$:2LcA	40	−3.12 (0.05)	0.45 (0.05)	2.41 (0.06)	4.73 (0.10)	0.37 (0.08)	0.48 (0.01)	38	0.99	1601
		50	−3.14 (0.05)	0.44 (0.05)	2.35 (0.06)	4.49 (0.09)	0.35 (0.08)	0.45 (0.01)	38	0.99	1709
		60	−3.18 (0.04)	0.43 (0.05)	2.29 (0.06)	4.25 (0.09)	0.36 (0.07)	0.42 (0.01)	38	0.99	1730
13	$[N_{2222}^+][Cl^-]$:2LvA	40	−3.30 (0.08)	0.13 (0.08)	2.44 (0.10)	4.80 (0.16)	−0.19 (0.13)	0.59 (0.02)	38	0.99	604
		50	−3.38 (0.09)	0.12 (0.09)	2.40 (0.11)	4.61 (0.17)	−0.18 (0.14)	0.56 (0.02)	38	0.99	476
		60	−3.34 (0.07)	0.16 (0.07)	2.28 (0.09)	4.36 (0.14)	−0.20 (0.12)	0.52 (0.02)	38	0.99	644
	$[N_{4444}^+][Cl^-]$	40[e]	−2.33 (0.10)	0.22 (0.10)	1.14 (0.14)	4.81 (0.19)	0 (0)	0.66 (0.03)	26	0.99	372
		50	−3.24 (0.12)	−0.17 (0.11)	2.72 (0.15)	8.03 (0.22)	−0.53 (0.18)	0.71 (0.03)	35	0.99	501
		60	−3.24 (0.10)	0 (0)	2.60 (0.13)	7.70 (0.19)	−0.44 (0.16)	0.67 (0.02)	39	0.99	609

[a] $[N_{4444}^+][Cl^-]$ IL is included for comparison purposes. [b] *n*, number of probe analytes subjected to multiple linear regression. [c] R^2, correlation coefficient. [d] *F*, Fisher *F*-statistic. [e] At 40 °C, the stationary phase becomes a solid resulting in a significant decrease in retention of analytes due to a prevailing gas−solid chromatography mechanism. System constants are observed to deviate from the two other temperatures due to the different separation mechanisms. The melting point of $[N_{4444}^+][Cl^-]$ is reported to be 41 and 52−54 °C in two different studies.[57,58]

Effect of the Hydrogen Bond Acceptor. In the case of ammonium-based DESs, four HBAs ($[N_{4444}^+][Cl^-]$, $[N_{4444}^+][Br^-]$, $[N_{3333}^+][Cl^-]$, and $[N_{2222}^+][Cl^-]$) were evaluated to study the effect that the length of alkyl chain substituent as well as anion have on individual solvation interactions. The hydrogen bond basicity (*a*-term) for the $[N_{4444}^+][Cl^-]$ IL was 8.03 at 50 °C and was higher than that of $[P_{66614}^+][Cl^-]$ (see Tables 1 and 2). This is consistent with previously reported system constants for these two ILs.[45,51] When the length of alkyl chain substituent in the cation was varied while maintaining the chloride anion with the same HBD, the hydrogen bond basicity of $[N_{4444}^+][Cl^-]$:2OA, $[N_{3333}^+][Cl^-]$:2OA, and $[N_{2222}^+][Cl^-]$:2OA decreased from (6.80 ± 0.14), (5.43 ± 0.14), and (5.02 ± 0.11) at 50 °C, respectively. A similar trend of decreasing hydrogen bond basicity was observed in the study of Teles et al., where the hydrogen bond basicity (*β*-parameter) of DESs decreased in order of 1.19, 0.96, and 0.87 for the $[N_{4444}^+][Cl^-]$:2OA, $[N_{3333}^+][Cl^-]$:2OA, and $[N_{2222}^+][Cl^-]$:2OA DESs, respectively.[31] When the chloride anion in the $[N_{4444}^+][Cl^-]$:2OA DES was changed to $[Br^-]$ in the $[N_{4444}^+][Br^-]$:2OA DES, the hydrogen bond basicity and dipolarity were both observed to decrease by nearly 20 and 10%, respectively, due to stronger hydrogen bonding interactions offered by the chloride anion.[31,45] Teles

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Table 2. System Constants for Seven Phosphonium-Based DESs Obtained from the Solvation Parameter Model[a]

			system constants								
no.	DES	temp. (°C)	c	e	s	a	b	l	n[b]	R^2[c]	F[d]
14	$[P_{66614}^+][Cl^-]$:OA	40	−3.10 (0.08)	−0.28 (0.08)	2.09 (0.10)	7.20 (0.15)	−0.56 (0.13)	0.85 (0.02)	39	0.99	952
		50	−3.19 (0.08)	−0.22 (0.07)	1.98 (0.09)	6.92 (0.14)	−0.44 (0.12)	0.81 (0.02)	38	0.99	924
		60	−3.12 (0.08)	−0.15 (0.08)	1.79 (0.10)	6.51 (0.15)	−0.45 (0.12)	0.76 (0.02)	38	0.99	847
15	$[P_{66614}^+][Cl^-]$:2OA	40	−3.05 (0.10)	−0.23 (0.09)	2.00 (0.12)	6.98 (0.18)	−0.57 (0.14)	0.82 (0.02)	37	0.99	571
		50	−3.16 (0.08)	−0.20 (0.07)	1.92 (0.10)	6.83 (0.15)	−0.50 (0.11)	0.81 (0.02)	37	0.99	825
		60	−3.12 (0.08)	−0.18 (0.07)	1.83 (0.09)	6.45 (0.14)	−0.48 (0.11)	0.77 (0.02)	37	0.99	812
16	$[P_{66614}^+][Cl^-]$:BSA	40	−2.98 (0.07)	−0.46 (0.07)	2.04 (0.08)	5.34 (0.12)	−0.68 (0.10)	0.86 (0.02)	41	0.99	1077
		50	−2.98 (0.06)	−0.43 (0.06)	1.97 (0.08)	5.04 (0.11)	−0.67 (0.10)	0.82 (0.02)	41	0.99	1181
		60	−2.97 (0.06)	−0.34 (0.06)	1.82 (0.07)	4.72 (0.10)	−0.59 (0.09)	0.77 (0.01)	41	0.99	1147
17	$[P_{66614}^+][Cl^-]$:2BSA	40	−3.01 (0.06)	−0.40 (0.06)	2.03 (0.08)	4.28 (0.13)	−0.39 (0.11)	0.83 (0.02)	39	0.99	917
		50	−3.07 (0.07)	−0.35 (0.06)	1.91 (0.08)	4.04 (0.12)	−0.39 (0.10)	0.80 (0.01)	40	0.99	1068
		60	−2.95 (0.05)	−0.26 (0.05)	1.72 (0.07)	3.56 (0.10)	−0.30 (0.09)	0.74 (0.01)	39	0.99	1112
18	$[P_{66614}^+][Cl^-]$:TSA	40	−3.23 (0.07)	−0.32 (0.07)	2.62 (0.09)	6.20 (0.15)	−0.58 (0.12)	0.77 (0.02)	37	0.99	1004
		50	−3.27 (0.07)	−0.30 (0.07)	2.56 (0.09)	5.96 (0.14)	−0.62 (0.12)	0.74 (0.02)	37	0.99	972
		60	−3.26 (0.06)	−0.33 (0.06)	2.51 (0.08)	5.63 (0.13)	−0.60 (0.11)	0.69 (0.01)	37	0.99	1015
19	$[P_{4444}^+][Cl^-]$:TSA	40	−3.18 (0.07)	−0.35 (0.07)	2.58 (0.10)	6.06 (0.14)	−0.56 (0.12)	0.78 (0.02)	37	0.99	915
		50	−3.25 (0.07)	−0.32 (0.07)	2.52 (0.10)	5.83 (0.14)	−0.57 (0.12)	0.74 (0.02)	37	0.99	855
		60	−3.17 (0.07)	−0.30 (0.07)	2.41 (0.09)	5.48 (0.13)	−0.60 (0.11)	0.69 (0.02)	37	0.99	864
20	$[P_{Al(Ph)3}^+][Br^-]$:3TSA	40	−3.29 (0.06)	0.27 (0.06)	1.91 (0.08)	5.12 (0.12)	0.70 (0.10)	0.67 (0.01)	36	0.99	1144
		50	−3.29 (0.06)	0.27 (0.06)	1.86 (0.07)	4.80 (0.11)	0.63 (0.09)	0.64 (0.01)	36	0.99	1160
		60	−3.32 (0.06)	0.28 (0.06)	1.82 (0.07)	4.49 (0.11)	0.59 (0.09)	0.60 (0.01)	36	0.99	1086
	$[P_{66614}^+][Cl^-]$	40	−2.92 (0.09)	−0.16 (0.08)	2.11 (0.11)	7.72 (0.17)	−0.69 (0.14)	0.80 (0.02)	37	0.99	814
		50	−3.07 (0.09)	−0.26 (0.08)	2.06 (0.10)	7.36 (0.15)	−0.59 (0.13)	0.81 (0.02)	39	0.99	845
		60	−3.12 (0.07)	−0.22 (0.07)	1.99 (0.10)	6.94 (0.14)	−0.54 (0.12)	0.77 (0.02)	40	0.99	970

[a] $[P_{66614}^+][Cl^-]$ IL is included for comparison purposes. [b] n, number of probe analytes subjected to multiple linear regression. [c] R^2, correlation coefficient. [d] F, Fisher F-statistic.

et al. have also reported a drop in the hydrogen bond basicity of DESs when the anion of $[N_{4444}^+][Cl^-]$:2OA DES (β = 1.19) was changed to $[Br^-]$ (β = 1.09).[31] The ability to tune the hydrogen bonding capability of DESs could be highly beneficial in the solubilization and stabilization of viruses.[52] The dipolarity of DESs with a 1:2 ratio of HBA to octanoic acid was observed to increase gradually from 1.97 ± 0.07 (DES **11**), 2.05 ± 0.09 (DES **10**), 2.15 ± 0.09 (DES **8**), 2.34 ± 0.10 (DES **2**) at 50 °C when comparing the $[N_{2222}^+][Cl^-]$, $[N_{3333}^+][Cl^-]$, $[N_{4444}^+][Br^-]$, $[N_{4444}^+][Cl^-]$ HBAs, respectively. Overall, dispersion interactions did not vary significantly when the structure of the HBA was varied.

Three different HBAs (i.e., $[P_{66614}^+][Cl^-]$, $[P_{4444}^+][Cl^-]$, $[P_{Al(Ph)3}^+][Br^-]$) were used to prepare the phosphonium-based DESs in this study. A comparison of the hydrogen bond basicity and dipolarity for the $[P_{66614}^+][Cl^-]$:TSA and $[P_{4444}^+][Cl^-]$:TSA DESs reveals that the HBA had very little effect on these two solvation interactions. The most dramatic effect was observed for the $[P_{Al(Ph)3}^+][Br^-]$:3TSA DES, which contains one allyl and three phenyl substituents appended to the phosphonium cation. The presence of aromatic moieties in the HBA and HBD gave rise to enhanced $\pi-\pi$ interactions and an "e" term of 0.27 at 50 °C. Dispersion interactions were similar for the $[P_{66614}^+][Cl^-]$:TSA and $[P_{4444}^+][Cl^-]$:TSA DESs but were considerably smaller for the $[P_{Al(Ph)3}^+][Br^-]$:3TSA DES.

Effect of Hydrogen Bond Donor. Seven different HBDs possessing a broad range of pK_a values were evaluated in the ammonium-based DESs and paired with HBAs at different molar ratios. For DESs formed using the $[N_{4444}^+][Cl^-]$ HBA, the hydrogen bond basicity decreases significantly when combined with lower pK_a acids. The neat $[N_{4444}^+][Cl^-]$ IL afforded the highest hydrogen bond basicity of 8.03 at 50 °C. Combining the $[N_{4444}^+][Cl^-]$ HBA with 1 molar equiv of octanoic acid (pK_a = 4.86) resulted in a drop to (7.60 ± 0.16) for the $[N_{4444}^+][Cl^-]$:OA DES. When stronger acids, such as *para*-toluenesulfonic acid (pK_a = −2.8) and benzenesulfonic acid (pK_a = −6.7), were employed as HBDs, the hydrogen bond basicity underwent another significant decrease to (5.66 ± 0.13) for $[N_{4444}^+][Cl^-]$:TSA and (5.23 ± 0.11) for $[N_{4444}^+][Cl^-]$:BSA at 50 °C, respectively, as shown in Figure 4a. Examining the effect of DESs formed using a 1:2 molar ratio of HBA/HBD, Figure 4b shows that the hydrogen bond basicity dropped by nearly 10 and 18% when comparing $[N_{4444}^+][Cl^-]$:OA/$[N_{4444}^+][Cl^-]$:2OA and $[N_{4444}^+][Cl^-]$:BSA/$[N_{4444}^+][Cl^-]$:2BSA DESs, respectively. Malonic acid, a dicarboxylic acid (pK_{a1} = 2.83), and lactic acid (pK_a = 3.86) were particularly effective at lowering the hydrogen bond basicity in the case of the $[N_{4444}^+][Cl^-]$:2MA (4.41 ± 0.10) and $[N_{4444}^+][Cl^-]$:2LcA (4.64 ± 0.10) DESs compared to that of the $[N_{4444}^+][Cl^-]$:2OA (6.80 ± 0.14) DES at 50 °C. This data is further illustrated in Figure 2a,b, where alcohols are observed to retain significantly longer on the $[N_{4444}^+][Cl^-]$:OA DES compared to the $[N_{4444}^+][Cl^-]$:2OA DES due to stronger hydrogen bond basicity interactions. In addition, it can also be observed that chlorohexane and chlorooctane interact more strongly with the $[N_{4444}^+][Cl^-]$:2OA DES due to higher dispersive-type interactions. These separations are contrasted by the $[N_{4444}^+][Cl^-]$:2LcA DES in Figure 2c, where alcohols exhibit very low retention (less than 13 min) and different separation selectivity due to lower hydrogen bond basicity and dispersion interactions. The trend of decreasing hydrogen bond basicity upon shortening the alkyl chain length of HBD in the $[N_{4444}^+][Cl^-]$:2OA (6.80 ± 0.14 at 50 °C) and $[N_{4444}^+][Cl^-]$:2HA (6.44 ± 0.15 at 50 °C) DESs was observed to be consistent with the Kamlet−Taft

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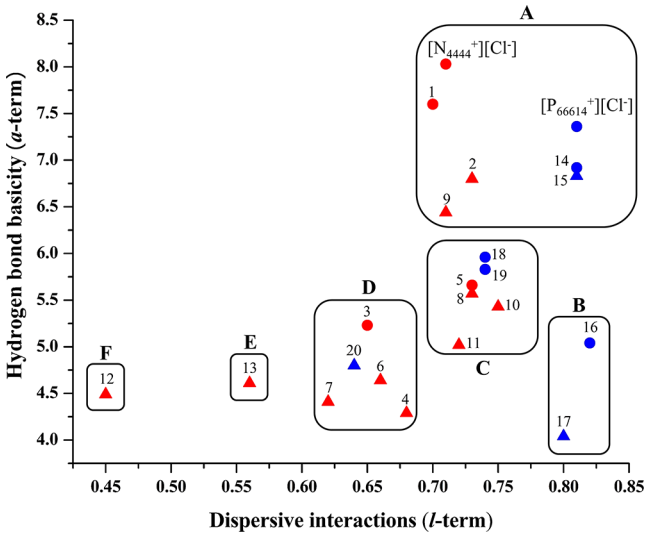


Figure 3. Grouping of 20 DESs based on hydrogen bond basicity and dispersive interactions using *k*-means clustering. For comparison and benchmarking purposes, the $[N_{4444}^+][Cl^-]$ and $[P_{66614}^+][Cl^-]$ ILs are also included. Red circles represent the ammonium-based DESs with HBA/HBD of 1:1 and neat $[N_{4444}^+][Cl^-]$ IL, and red triangles represent the ammonium-based DESs with HBA/HBD of 1:2. Blue circles represent the phosphonium-based DESs with HBA/HBD of 1:1 and neat $[P_{66614}^+][Cl^-]$ IL, while blue triangles represent the phosphonium-based DESs with HBA/HBD of 1:2 and 1:3 HBA/HBD for DES 20. **1**, $[N_{4444}^+][Cl^-]$:OA; **2**, $[N_{4444}^+][Cl^-]$:2OA; **3**, $[N_{4444}^+][Cl^-]$:BSA; **4**, $[N_{4444}^+][Cl^-]$:2BSA; **5**, $[N_{4444}^+][Cl^-]$:TSA; **6**, $[N_{4444}^+][Cl^-]$:2LcA; **7**, $[N_{4444}^+][Cl^-]$:2MA; **8**, $[N_{4444}^+][Br^-]$:2OA; **9**, $[N_{4444}^+][Cl^-]$:2HA; **10**, $[N_{3333}^+][Cl^-]$:2OA; **11**, $[N_{2222}^+][Cl^-]$:2OA; **12**, $[N_{2222}^+][Cl^-]$:2LcA; **13**, $[N_{2222}^+][Cl^-]$:2LvA; **14**, $[P_{66614}^+][Cl^-]$:OA; **15**, $[P_{66614}^+][Cl^-]$:2OA; **16**, $[P_{66614}^+][Cl^-]$:BSA; **17**, $[P_{66614}^+][Cl^-]$:2BSA; **18**, $[P_{66614}^+][Cl^-]$:TSA; **19**, $[P_{4444}^+][Cl^-]$:TSA; and **20**, $[P_{Al(Ph)3}^+][Br^-]$:3TSA.

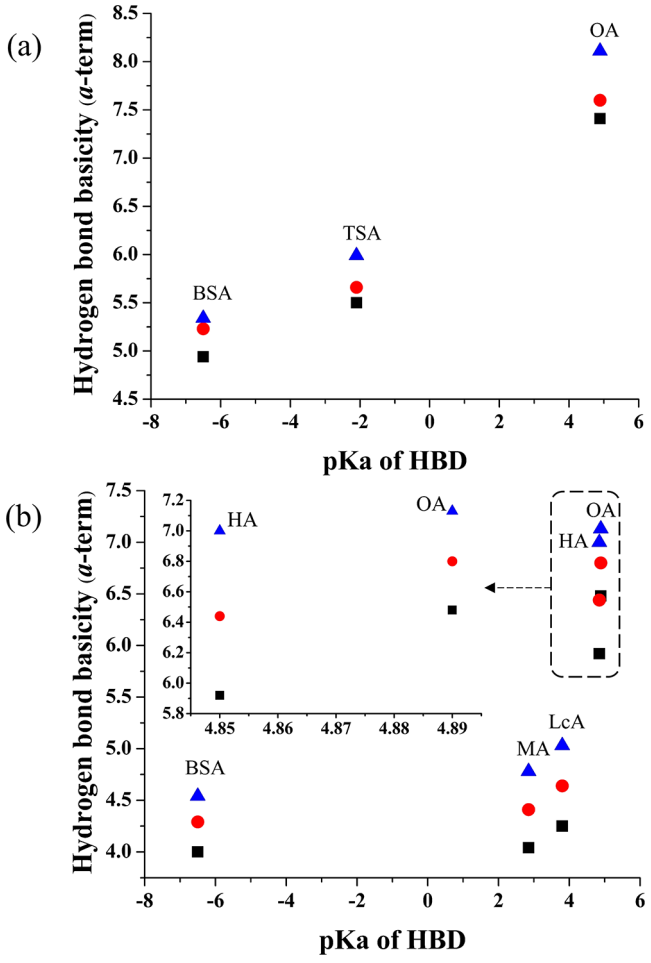


Figure 4. Plot illustrating the change in DES hydrogen bond basicity (*a*-term) upon changing the pK_a of HBD at temperatures of 40 (blue solid triangle), 50 (red solid circle), and 60 °C (solid square). Panel (a) shows the $[N_{4444}^+][Cl^-]$:BSA, $[N_{4444}^+][Cl^-]$:TSA, and $[N_{4444}^+][Cl^-]$:OA DESs with HBA/HBD of 1:1. Panel (b) plots the $[N_{4444}^+][Cl^-]$:2BSA, $[N_{4444}^+][Cl^-]$:2MA, $[N_{4444}^+][Cl^-]$:2LcA, $[N_{4444}^+][Cl^-]$:2HA, and $[N_{4444}^+][Cl^-]$:2OA DESs with HBA/HBD of 1:2. The inset within panel (b) eliminates overlap of $[N_{4444}^+][Cl^-]$:2OA and $[N_{4444}^+][Cl^-]$:2HA by plotting them on a different scale for better visualization.

hydrogen bond basicities, where a hydrogen bond basicity (*β*-parameter) of 1.19 and 1.02 was reported for the $[N_{4444}^+][Cl^-]$:2OA DES and $[N_{4444}^+][Cl^-]$:2HA DESs, respectively.[31] Phosphonium-based DESs consisting of three different HBDs (octanoic acid, benzenesulfonic acid, and *para*-toluenesulfonic acid) exhibited similar trends observed previously for the ammonium-based DESs. The neat $[P_{66614}^+][Cl^-]$ IL possessed the highest hydrogen bond basicity of (7.36 ± 0.15) at 50 °C. The hydrogen bond basicity dropped to (6.92 ± 0.14) for $[P_{66614}^+][Cl^-]$:OA, (5.96 ± 0.14) for $[P_{66614}^+][Cl^-]$:TSA, and (5.04 ± 0.11) for $[P_{66614}^+][Cl^-]$:BSA DES. A comparison of $[P_{66614}^+][Cl^-]$:OA and $[P_{66614}^+][Cl^-]$:2OA as well as $[P_{66614}^+][Cl^-]$:BSA and $[P_{66614}^+][Cl^-]$:2BSA reveals that 2 molar equiv of the HBD resulted in compounds with lower hydrogen bond basicity, with the effect being more pronounced for benzenesulfonic acid. Dispersion interactions were largely unchanged when using different molar equivalents of HBD while maintaining the same HBA. Retention factors for 15 probe molecules on 13 DES stationary phases are provided in Table S4 and allow for a comparison among DESs belonging to the two different groups.

To further understand the hydrogen bond acidic nature of the DESs, chromatographic retention of *N*,*N*-DMF and *N*,*N*-dimethylacetamide (*N*,*N*-DMAC) was studied on six ammonium-based DES stationary phases with 1:1 and 1:2 molar ratios of HBD/HBA. These probes were chosen due to the fact that they are sufficiently volatile and their retention should vary depending upon the acidity of the DES (e.g., the more acidic the DES results in increased retention of basic probes). Considering the $[N_{4444}^+][Cl^-]$:OA, $[N_{4444}^+][Cl^-]$:TSA, and $[N_{4444}^+][Cl^-]$:BSA DESs, the retention factors of *N*,*N*-DMF and *N*,*N*-DMAC increased significantly moving from OA (15.27 and 17.40, respectively) to the more acidic BSA HBD (19.50 and 42.17), indicating an enhancement in hydrogen bond acidity interactions as the hydrogen bond basicity of the DES decreases (Table 3). The retention factors on the $[N_{4444}^+][Cl^-]$:OA and $[N_{4444}^+][Cl^-]$:2OA DESs were unchanged for *N*,*N*-DMF and slightly higher for *N*,*N*-DMAC on the $[N_{4444}^+][Cl^-]$:2OA DES. The difference was much more pronounced for the $[N_{4444}^+][Cl^-]$:BSA and $[N_{4444}^+][Cl^-]$:2BSA DESs, where the retention factor for *N*,*N*-DMAC increased dramatically from 42.17 to 1611.42. Clearly, the hydrogen bond-donating capability of DESs in this class can be enhanced using low pK_a HBDs and employing higher molar ratios of HBD to HBA.

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Table 3. Retention Factors of *N,N*-Dimethylformamide and *N,N*-Dimethylacetamide on Ammonium-Based DESs Possessing Various Acidic HBDs at 60 °C[a]

	neat IL	HBA/HBD (1:1)				HBA/HBD (1:2)			
probe	$[N_{4444}^+][Cl^-]$	$[N_{4444}^+][Cl^-]$:OA	$[N_{4444}^+][Cl^-]$:TSA	$[N_{4444}^+][Cl^-]$:BSA	$[N_{4444}^+][Cl^-]$:2OA	$[N_{4444}^+][Cl^-]$:2LcA	$[N_{4444}^+][Cl^-]$:2BSA	
N,N-dimethylformamide (*N,N*-DMF)	26.21	15.27	17.05	19.50	15.87	30.50	49.68	
N,N-dimethylacetamide (*N,N*-DMAC)	27.38	17.40	22.85	42.17	20.67	36.38	1611.42	

[a]For clarity purposes, DESs comprised of 1:1 and 1:2 molar ratio of HBA/HBD are separated from one another. The $[N_{4444}^+][Cl^-]$ IL is included for comparison purposes. Note: retention factor (k) is determined by the following equation: $k = (t_R - t_0/t_0)$, where t_R is the retention time of the probe and t_0 is the dead time of an unretained molecule, propane.

DES Solvation Model. DESs have been increasingly applied as green solvents in chemical separations since their multiple solvation interactions can be tuned compared to conventional organic solvents.[53] The system constants determined for the DESs in this study provide insight into the magnitude of individual solvation interactions and how they can be tailored using different HBA and HBDs. This section will focus on using the solvation data from this study to understand qualitative observations and results from previously published studies involving DESs.

Desulfurization of Fuels with DESs. DESs have been used as efficient extraction solvents for the desulfurization of fuels.[22−24,54] Organosulfur compounds are a major source of SO_x produced from fuel combustion and contribute significantly toward environmental pollution.[24] DESs consisting of ammonium halide-based HBAs and carboxylic acid-based HBDs have been shown to exhibit high extraction efficiencies in the extractive desulfurization of fuels.[22−25,55] Li et al. reported an approximate 25% decrease in extraction efficiency moving from the $[N_{4444}^+][Cl^-]$:2OA to $[N_{4444}^+][Cl^-]$:2MA DES. It was concluded that the main driving force for the desulfurization process is hydrogen bonding between the DES and the thiol functional group of sulfur-containing compounds in fuels.[22] Table 1 shows a considerable difference in hydrogen bond basicity between $[N_{4444}^+][Cl^-]$:2OA (6.80 ± 0.14) and $[N_{4444}^+][Cl^-]$:2MA (4.41 ± 0.10) with both DESs falling into different groups within Figure 3. It has also been shown that extraction efficiencies further decreased when the length of alkyl chain substituent within the HBA was reduced from $[N_{4444}^+][Cl^-]$ to tetramethylammonium chloride $[N_{1111}^+][Cl^-]$.[22] A similar trend can be observed in Table 1 when the HBA is varied from $[N_{4444}^+][Cl^-]$ (6.80 ± 0.14), $[N_{3333}^+][Cl^-]$ (5.43 ± 0.14), and $[N_{2222}^+][Cl^-]$ (5.02 ± 0.11) at 50 °C with octanoic acid as HBD. As shorter alkyl substituents are incorporated within the HBA or the number of carboxyl or hydroxyl functional groups within the HBD is increased, the capability of the DES to undergo hydrogen bonding interactions with organosulfur compounds decreases.

Extraction of Natural Products by DESs. DESs containing tetraalkylammonium halide-based HBAs and carboxylic acid-based HBDs have been used for the extraction of cynaropicrin, a bioactive compound of great potential for its applications in nutraceuticals.[40,56] The hydroxyl groups within the chemical structure of cynaropicrin make it especially prone to undergo hydrogen bonding interactions with the class of DESs examined in this study. An extraction efficiency of 2.4% has been reported using $[N_{4444}^+][Cl^-]$:2HA as extraction solvent, while a 0.4% extraction efficiency was observed when the HBA was shortened to the $[N_{2222}^+][Cl^-]$:2HA DES.[40] This observation is supported by the results in the solvation model, where decreasing the alkyl chain substituent from butyl to ethyl within the cation of HBA resulted in a significant drop in the capability of DES to undergo hydrogen bonding interactions with solute molecules (see Table 1). Figure S4 shows the extraction yield (blue bar) of cynaropicrin using DESs $[N_{4444}^+][Cl^-]$:2OA, $[N_{4444}^+][Cl^-]$:2HA, and $[N_{4444}^+][Br^-]$:2OA as well as the hydrogen bond basicity values (red bar) determined in this study at 40 °C for the same DESs. The system constants reveal a direct relation to the hydrogen bond basicity of the solvent and the extraction yield of cynaropicrin. When the chloride anion was replaced with bromide while maintaining the same HBD, as in the case of $[N_{4444}^+][Cl^-]$:2OA and $[N_{4444}^+][Br^-]$:2OA, the extraction yield

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decreased from 2.7 to 1.7%.[40] The hydrogen bonding basicity values determined for the $[N_{4444}^+][Cl^-]$:2OA and $[N_{4444}^+][Br^-]$:2OA DESs at 40 °C were (7.13 ± 0.15) and (5.88 ± 0.13), respectively.

CONCLUSIONS

Deep eutectic solvents (DESs) have demonstrated great potential as sustainable solvents due to the fact that they generally involve cheap and environmentally benign starting materials. Because many of their properties can be tuned and modulated using different HBA and HBD combinations, they are among the most complex solvent systems and can undergo a multitude of different interactions. Single parameter polarity scales often provide a weighted average of all solute−solvent interactions and do not distinguish between dominant and less influential interactions. The solvation parameter model employs chromatographic retention data obtained for many different probe molecules to determine individual solvation interactions of DESs possessing different HBA/HBD composition.

For DESs evaluated in this study, the hydrogen bond basicity can be modulated by (1) choosing more acidic (lower pK_a) carboxylic acids as HBDs and using larger molar ratios of HBD compared to the HBA, (2) decreasing the length of alkyl chain substituents within HBA, and (3) varying the halide anion within the HBA. Using highly basic probes to examine the acidity of DESs, it was found that DESs formed with larger molar ratios of low pK_a HBAs result in solvents with higher hydrogen bond acidity. Dispersive interactions were found to be higher for the phosphonium-based DESs and were heavily influenced by the type of HBD as well as the molar ratio. Based upon hydrogen bond basicity and dispersion interactions, DESs can be clustered into six groups. Classifying DESs in terms of their multiple solvation interactions aids in their rational design and enhances our understanding of the complex interplay between the HBA and HBD.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssuschemeng.0c06926.

Water content of DESs, melting point/glass transition temperatures of DESs, list of probe molecules and corresponding solute descriptors, retention factors of probe molecules on ammonium and phosphonium-based DES stationary phases, clustering of DESs using *k*-means clustering algorithm, and ^1H and ^{13}C NMR spectra for DESs evaluated in the study (PDF)

AUTHOR INFORMATION

Corresponding Author

Jared L. Anderson − *Ames Laboratory−USDOE and Department of Chemistry, Iowa State University, Ames, Iowa 50011, United States;* orcid.org/0000-0001-6915-8752; Phone: +1 515-294-8356; Email: andersoj@iastate.edu

Authors

Muhammad Qamar Farooq − *Ames Laboratory−USDOE and Department of Chemistry, Iowa State University, Ames, Iowa 50011, United States*

Gabriel A. Odugbesi − *Department of Chemistry, Iowa State University, Ames, Iowa 50011, United States*

Nabeel Mujtaba Abbasi − *Ames Laboratory−USDOE and Department of Chemistry, Iowa State University, Ames, Iowa 50011, United States*

Complete contact information is available at: https://pubs.acs.org/10.1021/acssuschemeng.0c06926

Notes

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